

IFU Proofreading Report

AUDIT TRAIL · CHANGE CONTROL RECORD

REPORT ID	GENERATED	USER	TOTAL DOCUMENTS
	2026-08-25	.	1 pair

Inspection Summary

Case	Change	Deletion	Insertion	Space
15	43	1	20	3

Change Log

ID	Change Type	Current IFU	Revised IFU	Comments
1 p1	Change	• Complete each procedure: Preparation for Cleaning, Manual Cleaning or Automated Cleaning, Disinfection and Sterilization when reprocessing the instruments.	• Complete each procedure: Preparation preparation for Cleaning, cleaning, Manual manual Cleaning cleaning or Automated automated Cleaning, cleaning, Disinfection disinfection, and Sterilization sterilization when reprocessing the instruments.	
2 p1	Change	Wear appropriate protective equipment (gloves, eye protection, etc.) when reprocessing any medical device.	Wear appropriate protective equipment (gloves, eye eye protection, etc.) when reprocessing any medical device.	
3 p1	Change	Point of Use Care The drying of gross soil (blood, tissue and/or debris) on devices following surgical use should be avoided.	Point of Use Care The drying of gross soil (blood, tissue and/or debris) on devices following surgical use should be avoided.	
4 p1	Change	Do not use saline, environmental disinfection (including chlorine solutions) or surgical antiseptics (such as iodine- or chlorhexidine-containing products.)	• Do not use saline, environmental disinfection (including chlorine solutions) or surgical antiseptics (such as iodine- or chlorhexidine-containing products, products).	
5 p1	Change	Flush all lumens, blind holes small clearances, and moving and intricate parts with water (or detergent solution) to prevent the drying of soil and/or debris.	Flush all lumens, blind holes holes, small clearances, and moving and intricate parts with water (or detergent solution) to prevent the drying of soil and/or debris.	
6 p1	Change	Transport to Processing Area Surgically used devices may be considered biohazardous and should be safely transported to a designated processing area in accordance with local policies.	Transport to Processing processing Area area Surgically used devices may be considered biohazardous bio-hazardous and should be safely transported to a designated processing area in accordance with local policies.	
7 p1	Change	Preparation for Cleaning It is recommended that devices should be reprocessed as soon as is reasonably practical following surgical use.	Preparation for Cleaning It is recommended that devices should be reprocessed as soon as is reasonably practical following surgical use.	
8 p1	Change	• Care should be taken in the handling and cleaning of sharp devices.	• Table Care 2: should be taken in the handling and Automatic cleaning of sharp devices, parameters.	

ID	Change Type	Current IFU	Revised IFU	Comments
9 p1	Change	These are recommended to be cleaned separately to reduce risks of injury.	• Care should be taken in the handling and cleaning of sharp devices. These are recommended to be cleaned separately to reduce risks of injury injury.	
10 p1	Change	• Disassemble the instruments.	• Disassemble the instruments instruments according to the following instructions:	
11 p1	Space	1. Disconnect the light cable from the light guidepost.	1. Disconnect the light cable from the light lightguide guidepost post.	
12 p1	Deletion	2. Disconnect the camera-coupling device from the eyepiece	—	
13 p1	Change	3. Remove any light cable adapters from the endoscope.	3. Remove any light cable adapters adaptors from the endoscope.	
14 p1	Change	3. Remove any light cable adapters from the endoscope.	3.2. Remove Sterilize any the endoscope and light cable adapters adaptors from the endoscope. disassembled.	
15 p1	Change	4. Disassemble the endoscope from any devices or accessories used during the procedure.	4. Disassemble the endoscope from any devices or accessories compatible devices used during the procedure.	
16 p1	Change	5. Disassemble the sheath including the stopcocks according to Figure 2: Disassembly	5. Disassemble the sheath including the stopcocks according to Figure 2: Disassembly	
17 p1	Change	6. Perform either the Manual Cleaning Procedure or the Automated Cleaning procedure below.	6. Perform either the Manual manual Cleaning cleaning Procedure procedure or the Automated automated Cleaning cleaning procedure described below.	
18 p1	Change	MANUAL CLEANING Perform the following steps at the point of central reprocessing	MANUAL CLEANING Perform the following steps at the point of central reprocessing reprocessing:	
19 p1	Change	4. Soak for five (5) minutes and then use soft bristle brushes to remove any debris/soil from the instruments while they are submerged.	4. Soak for five (5) minutes and then use soft bristle brushes to remove any debris/soil debris or soil from the instruments while they are submerged.	
20 p1	Change	Ensure that the brushes can access the hard-to-reach areas such as cannulations, cracks, crevices, and threads.	Ensure that the brushes can access the hard-to-reach areas such as cannulations, cracks, crevices, and threads threads. Remove all visible debris.	
21 p1	Change	Remove all visible debris.	Remove all any visible excess debris lubricant.	
22 p1	Change	5. Fill a syringe with the detergent and use to flush	5. Fill 5. Fill a syringe with the detergent and use to flush cannulations, crevices, and threads while submerged.	
23 p1	Case	6. Repeat the Brushing step (Step	6. Repeat the Brushing brushing step (Step (step	
24 p1	Case	3) and the Flushing step (Step	3) and the Flushing flushing step (Step (step	
25 p1	Change	8. For the Sheath only: Completely submerge the devices in an ultrasonic bath.	8. For the Sheath sheath only only, Completely completely submerge the devices in an ultrasonic bath.	

ID	Change Type	Current IFU	Revised IFU	Comments
26 p1	Change	Carry out an ultrasonic second soaking bath in a neutral pH detergent solution for 10 minutes using an ultrasonic bath.	Carry out an ultrasonic second soaking bath in a neutral pH detergent solution for 10 minutes using an ultrasonic bath bath (refer to cleaning agent solution manufacturer's instructions for immersion time and temperature).	
27 p1	Change	(Refer to cleaning agent solution manufacturer's instructions for immersion time and temperature.)	(Refer to a cleaning agent lubricant solution in accordance with the lubricant manufacturer's instructions for immersion lubricate time parts and as temperature.) necessary. Use a cleaning water-soluble agent lubricant solution in accordance with the lubricant manufacturer's instructions for immersion lubricate time parts and as temperature.) necessary.	
28 p1	Change	10. For the final rinsing processes, soak the devices for a minimum of three (3) minutes in purified (e.g., deionized, reverse osmosis, distilled, or Critical Water), per AAMI TIR 34:2014, Dry with a lint-free soft cloth.	10. For the final rinsing processes, soak the devices for a minimum of three (3) minutes in purified (e.g., deionized, reverse osmosis, distilled, or Critical critical Water water), per AAMI TIR 34:2014 , Dry with a lint-free soft cloth.	
29 p1	Insertion	—	CLEANING, THERMAL DISINFECTION, AND STERILIZATION These cleaning, disinfection, and sterilization instructions have been validated for preparing reusable DePuy Synthes instruments for reuse.	
30 p1	Insertion	—	It is the responsibility of the end user to ensure that the cleaning, disinfection, and sterilization is performed using appropriate equipment, materials, and personnel to achieve the desired result.	
31 p1	Insertion	—	Any deviation from these instructions should be evaluated for effectiveness and potential adverse consequences.	
32 p1	Insertion	—	General considerations for processing	
33 p1	Insertion	—	• Observe point of use and transport procedures as described below.	
34 p1	Insertion	—	• Highly alkaline conditions (pH>10) can damage components or device.	
35 p1	Insertion	—	Point of use care	
36 p1	Insertion	—	Preparation for cleaning	
37 p1	Insertion	—	2.	
38 p1	Insertion	—	4	
39 p2	Change	AUTOMATED CLEANING Equipment Required: Automated washing shall be conducted in a validated washer-disinfector in compliance to ISO 15883-1 and -2, or to an equivalent standard.	AUTOMATED CLEANING Equipment Required required: Automated washing shall be conducted in a validated washer-disinfector in compliance to ISO 15883-1 and -2 and-2, or to an equivalent standard.	
40 p2	Change	2. Place the endoscope and its light cable adapters into a separate mesh basket.	2. Place the endoscope and its light cable adapters adaptors into a separate mesh basket.	

ID	Change Type	Current IFU	Revised IFU	Comments
41 p2	Change	Ensure that the endoscope and light cable adapters do not touch each other when placed into the basket.	Ensure that the endoscope and light cable adapters adaptors do not touch each other when placed into the basket.	
42 p2	Change	Place a mesh screen over the basket to contain the endoscope and light cable adapters in the basket during processing.	Place a mesh screen over the basket to contain the endoscope and light cable adapters adaptors in the basket during processing.	
43 p2	Change	Clean in a validated washer disinfectant using the Clean in a Validated washer disinfectant using the "INSTRUMENTS" cycle and a pH neutral cleaning agent intended for use in automated cleaning as shown in Table	3. Clean in a validated washer-disinfectant using the Clean in a Validated washer disinfectant using the "INSTRUMENTS" cycle and a pH neutral cleaning agent intended for use in automated cleaning as shown in Table	
44 p2	Change	1. Note: Enzol (1 oz/gal) and Prolystica 2X Neutral (1/8 oz/gal) were used for the validation.	1-2. Note: NOTE: Enzol (1 oz/gal)Enzol®(1oz/gal) and Prolystica 2X NeutralNeutral® (1/8 oz/gal) were used for the validation.	
45 p2	Change	Table 1: Automatic Cleaning Parameters: INSPECTION AFTER CLEANING After either Manual Cleaning or Automated Cleaning, visually inspect the instruments to verify the absence of visible soil, stains and debris.	Table 1 1:- Automatic Cleaning Parameters: INSPECTION AFTER CLEANING After either Manualmanual Cleaningcleaning or Automatedautomated Cleaningcleaning, visually inspect the instruments to verify the absence of visible soil, stains stains, and debris.	
46 p2	Space	Thermal disinfection should be conducted in a washer-disinfectant compliant to ISO 15883-1 and -2, or to an equivalent standard.	Thermal disinfection should be conducted in a washer-disinfectant compliant to ISO 15883-1 and -2, and-2, or to an equivalent standard.	
47 p2	Change	If this is not possible due to space limitations within the washer-disinfectant, use an irrigating rack/load carrier with connections designed to ensure an adequate flow of process fluids to the lumen or cannulation of the device if provided.	If this is not possible due to space limitations within the washer-disinfectant, washer- disinfectant, use an irrigating rack/load rack or load carrier with connections designed to ensure an adequate flow of process fluids to the lumen or cannulation of the device if provided.	
48 p2	Change	The following automated cycles are examples of validated cycles: STERRAD Sterilization: Endoscope and adapters only CAUTION: The sheaths are not validated for sterilization using STERRAD.	The STERRAD™ followingsterilization: automated-cycles-are examples of validated cycles: STERRAD Sterilization: Endoscope endoscope and adapters light adaptors only CAUTION: The sheaths are not validated for sterilization using STERRAD. STERRAD	
49 p2	Change	1. Ensure the endoscope and light cable adapters are disassembled before sterilization.	1. Ensure the endoscope and light cable adapters adaptors are disassembled before sterilization.	
50 p2	Change	2. The endoscope and light cable adapters can be sterilized in the STERRAD 100S system, and in the STERRAD 100NX and STERRAD NX sterilization systems using the STANDARD Cycle.	2. The endoscope and light cable adapters adaptors can be sterilized in the STERRAD 100S STERRAD™ 100S system, and in the STERRAD 100NX STERRAD™ 100NX, and STERRAD NX STERRAD™ NX sterilization systems using the STANDARD Cycle. cycle.	
51 p2	Change	3. Clean and thoroughly dry the endoscope and light cable adapters per the Cleaning/Disinfection Section.	3. Clean and thoroughly dry the endoscope and light cable adapters adaptors per the Cleaning/Disinfection section Section: titled "CLEANING, THERMAL DISINFECTION, AND STERILIZATION."	

ID	Change Type	Current IFU	Revised IFU	Comments
52 p2	Change	4. Place the endoscope and light cable adapters into the appropriate location within the Mitek Sport Medicine sterilization tray that is compatible with the STERRAD 100S, NX and 100NX sterilization systems.	4. Place the endoscope and light cable adapters adaptors into the appropriate location within the Mitek Sport Medicine sterilization tray that is compatible with the STERRAD 100S, NX NX, and 100NX and 100NX sterilization systems.	
53 p2	Change	Package trays/ instruments with a barrier wrap material in accordance with local procedures, using standardized wrapping techniques such as those described in ANSI/AAMI	Package trays trays or instruments with a barrier wrap material in accordance with local procedures, using standardized wrapping techniques such as those described in ANSI/AAMI	
54 p2	Change	79. In the United States use an FDA-cleared sterilization wrap.	79. In the United States States, use an FDA-cleared sterilization wrap.	
55 p2	Insertion	—	NOTE: The cleaning solution may contain enzymes.	
56 p2	Insertion	—	34. Dry with a lint-free soft cloth.	
57 p3	Insertion	—	The following automated cycles are examples of validated cycles (Table 3): Table 3: Examples of automated validated cycles.	
58 p3	Insertion	—	Table 3: Examples of automated validated cycles.	
59 p3	Insertion	—	Autoclave (steam) sterilization CAUTION: Do not use an "immediate use or flash" cycle to sterilize the instruments.	
60 p3	Insertion	—	1. Before assembly of the sheath for sterilization, make sure that all O-rings are in place and are not damaged.	
61 p3	Insertion	—	Remove and replace any O-rings that show any sign of damage or wear.	
62 p3	Insertion	—	Sterilize the sheath with the stopcocks in the open position.	
63 p3	Insertion	—	3. If the instruments come with a protective transport sleeve, do not sterilize the endoscopic compatible devices with the protective transport sleeve.	
64 p3	Insertion	—	1	
65 p2 tbl r1c2	Case	Recirculation Time set points	Recirculation time set points	
66 p2 tbl r1c3	Case	Water Temperature	Water temperature	
67 p2 tbl r1c4	Case	Detergent Type	Detergent type	

ID	Change Type	Current IFU	Revised IFU	Comments
68 p2 tbl r2c1	Case	Pre-Wash	Pre-wash	
69 p2 tbl r2c4	Change	N/A	Not applicable	
70 p2 tbl r3c1	Case	Enzyme Wash	Enzyme wash	
71 p2 tbl r4c3	Case	65°C (Set Point)	65°C (set point)	
72 p2 tbl r5c4	Change	N/A	Not applicable	
73 p2 tbl r6c1	Case	Dry Phase	Dry phase	
74 p2 tbl r6c3	Case	66°C (Set Point)	66°C (set point)	
75 p2 tbl r6c4	Change	N/A	Not applicable	
76 p2 tbl r1c2	Case	Recirculation Time (mins)	Recirculation time (mins)	
77 p2 tbl r1c3	Case	Water Temp	Water temp	
78 p2 tbl r1c4	Case	Water Type	Water type	
79 p2 tbl r2c1	Case	Thermal Disinfection	Thermal disinfection	
80 p2 tbl r2c3	Space	>90° □ (194°F)	> 90° □ (194°F)	
81 p2 tbl r3c1	Case	Thermal Disinfection	Thermal disinfection	

ID	Change Type	Current IFU	Revised IFU	Comments
82 p2 tbl r3c3	Change	>90°C (194°F)	> 90□ (194°F)	

Annotated Comparison

Red = deleted characters (A) | Green = inserted characters (B)

Page 1

Current (A)	Revised (B)
Point of Use Care - The drying of gross soil (blood, tissue and/or debris) on devices following surgical use should be avoided. It is preferred that gross soil is removed from devices following use and in preparation for transportation to a processing area. Gross soil can be removed using sponges, cloths, or soft brushes. Water and/or cleaning detergents (labelled for use on medical devices) may be used. - Do not use saline, environmental disinfection (including chlorine solutions) or surgical antiseptics (such as iodine- or chlorhexidine-containing products.) Flush all lumens, blind holes, small clearances, and moving and intricate parts with water (or detergent solution) to prevent the drying of soil and/or debris. - If gross soil cannot be removed at the point of use, the devices should be transported to prevent drying (e.g., covered with a towel dampened with purified water) and cleaned as soon as possible at a designated processing area. Transport to Processing Area Surgically used devices may be considered biohazardous and should be safely transported to a designated processing area in accordance with local policies. Complete the cleaning procedures for the instruments within 30 minutes following their use. Preparation for Cleaning - It is recommended that devices should be reprocessed as soon as is reasonably practical following surgical use. - Instruments must be cleaned separately from instrument trays and cases. - Care should be taken in the handling and cleaning of sharp devices. These are recommended to be cleaned separately to reduce risks of injury. - Disassemble the instruments. - Disconnect the light cable from the light guidepost. - Disconnect the camera-coupling device from the eyepiece. - Remove any light cable adapters from the endoscope. - Disassemble the endoscope from any devices or accessories used during the procedure. - Disassemble the sheath including the stopcocks according to Figure 2: Disassembly - Perform either the Manual Cleaning Procedure or the Automated Cleaning procedure below. MANUAL CLEANING Perform the following steps at the point of central reprocessing: - Rinse the instruments under cold running tap water for a minimum of three (3) minutes. Brush all surfaces and actuate moving parts during the rinse. - Prepare a neutral (pH 7-9) detergent (cleaning) solution (which may or may not include enzymes) in accordance to the detergent manufacturer's instructions. Use a fresh mixture of cleaning solution for each set of instruments when cleaning. NOTE: The cleaning solution may contain enzymes. Aluminum-safe alkaline cleaners can be used but can vary in material compatibility overtime based on their formulation. Material compatibility should be confirmed with the detergent manufacturer. - Immerse the instruments completely in the detergent solution for a minimum of 15 minutes. - Soak for five (5) minutes and then use soft bristle brushes to remove any debris/soil from the instruments while they are submerged. Brush for a minimum of 3 minutes. Ensure that the brushes can access the hard-to-reach areas such as cannulations, cracks, crevices, and threads. Remove all visible debris. - Fill a syringe with the detergent and use to flush cannulations, crevices and threads while submerged. - Repeat the Brushing step (Step 3) and the Flushing step (Step 4) an additional 2 times. - Check for remaining debris and remove as needed. - For the Sheath only: Completely submerge the devices in an ultrasonic bath. Carry out an ultrasonic second soaking bath in a neutral pH detergent solution for 10 minutes using an ultrasonic bath. (Refer to cleaning agent solution manufacturer's instructions for immersion time and temperature.) NOTE: Ultrasonic cleaning is only effective if the surface to be cleaned is immersed in the cleaning solution. Air pockets will decrease the efficacy of ultrasonic cleaning. - Rinse the instruments under cold running tap water for at least three (3) minutes while brushing with a soft bristle brush during the rinse. - For the final rinsing processes, soak the devices for a minimum of three (3) minutes in purified (e.g., deionized, reverse osmosis, distilled, or Critical Water), per AAMI TIR 34:2014. Dry with a lint-free soft cloth.	validated for preparing reusable DePuy Synthes instruments for reuse. It is the responsibility of the end user to ensure that the cleaning, disinfection, and sterilization is performed using appropriate equipment, materials, and personnel to achieve the desired result. Any deviation from these instructions should be evaluated for effectiveness and potential adverse consequences. General considerations for processing - Observe point of use and transport procedures as described below. - Highly alkaline conditions (pH>10) can damage components or device. - Complete each procedure: preparation for cleaning, manual cleaning or automated cleaning, disinfection, and sterilization when reprocessing the instruments. - Personnel trained in the appropriate reprocessing techniques and safety should complete all reprocessing. Wear appropriate protective equipment (gloves, eye protection, etc.) when reprocessing any medical device. Point of use care - The drying of gross soil (blood, tissue and/or debris) on devices following surgical use should be avoided. It is preferred that gross soil is removed from devices following use and in preparation for transportation to a processing area. Gross soil can be removed using sponges, cloths, or soft brushes. Water and/or cleaning detergents (labelled for use on medical devices) may be used. - Do not use saline, environmental disinfection (including chlorine solutions) or surgical antiseptics (such as iodine- or chlorhexidine-containing products). Flush all lumens, blind holes, small clearances, and moving and intricate parts with water (or detergent solution) to prevent the drying of soil and/or debris. - If gross soil cannot be removed at the point of use, the devices should be transported to prevent drying (e.g., covered with a towel dampened with purified water) and cleaned as soon as possible at a designated processing area. Transport to processing area Surgically used devices may be considered bio-hazardous and should be safely transported to a designated processing area in accordance with local policies. Complete the cleaning procedures for the instruments within 30 minutes following their use. Preparation for cleaning - It is recommended that devices should be reprocessed as soon as is reasonably practical following surgical use. - Instruments must be cleaned separately from instrument trays and cases. - Care should be taken in the handling and cleaning of sharp devices. These are recommended to be cleaned separately to reduce risks of injury. - Disassemble the instruments according to the following instructions: - Disconnect the light cable from the lightguide post. - Disconnect the camera-coupling device from the eyepiece. - Remove any light cable adaptors from the endoscope. - Disassemble the endoscope from any devices or compatible devices used during the procedure. - Disassemble the sheath including the stopcocks according to Figure 2. - Perform either the manual cleaning procedure or the automated cleaning procedure described below. MANUAL CLEANING Perform the following steps at the point of central reprocessing: - Rinse the instruments under cold running tap water for a minimum of three (3) minutes. Brush all surfaces and actuate moving parts during the rinse.

Current (A)

Revised (B)

manufacturer's instructions.

1. Manually clean the instruments by immersion prior to automated cleaning (reference manual cleaning section).
2. Place the endoscope and its light cable adapters into a separate mesh basket. Ensure that the endoscope and light cable adapters do not touch each other when placed into the basket. Place a mesh screen over the basket to contain the endoscope and light cable adapters in the basket during processing.

Clean in a validated washer disinfecter using the "INSTRUMENTS" cycle and a pH neutral cleaning agent intended for use in automated cleaning as shown in Table 1. **Note:** EnzoI® (1 oz/gal) and Prolystica 2X Neutral (1/8 oz/gal) were used for the validation.

Table 1: Automatic Cleaning Parameters:

Phase	Recirculation Time set points	Water Temperature	Detergent Type
Pre-Wash	2 minutes	Cold tap water	N/A
Enzyme Wash	1 minute	Hot tap water	Enzymatic detergent
Wash 1	2 minutes	65°C (Set Point)	Neutral pH instrument cleaner
Rinse 1	15 seconds	Hot tap water	N/A
Dry Phase	15 minutes	66°C (Set Point)	N/A

INSPECTION AFTER CLEANING

After either Manual Cleaning or Automated Cleaning, visually inspect the instruments to verify the absence of visible soil, stains and debris. If soil is present after cleaning, repeat cleaning procedure.

Thermal Disinfection

Thermal disinfection is recommended to render devices safe for handling prior to steam sterilization. Thermal disinfection should be conducted in a washer-disinfecter compliant to ISO 15883-1 and -2, or to an equivalent standard. Thermal disinfection in the washer-disinfecter shall be validated to provide an A₀ of at least 600 (e.g., 90°C (194°F) for 1 min). Higher levels of A₀ can be achieved by increasing the exposure time and temperature (e.g., A₀ of 3000 at >90°C (194°F) for 5 min, in accordance with local requirements). Load the device components in the washer-disinfecter in accordance with manufacturer's instructions, ensuring that the devices and lumens can drain freely. Lumen devices should be placed in a vertical position. If this is not possible due to space limitations within the washer-disinfecter, use an irrigating rack/load carrier with connections designed to ensure an adequate flow of process fluids to the lumen or cannulation of the device if provided.

The following automated cycles are examples of validated cycles:

Phase	Recirculation Time (mins)	Water Temp	Water Type
Thermal Disinfection	1	>90°C (194°F)	Critical water (RO, deionized or distilled water)
Thermal Disinfection	5	>90°C (194°F)	Critical water (RO, deionized or distilled water)

STERRAD Sterilization: Endoscope and adapters only

CAUTION: The sheaths are not validated for sterilization using STERRAD.

1. Ensure the endoscope and light cable adapters are disassembled before sterilization.
2. The endoscope and light cable adapters can be sterilized in the STERRAD 100S system, and in the STERRAD 100NX and STERRAD NX sterilization systems using the STANDARD Cycle.
3. Clean and thoroughly dry the endoscope and light cable adapters per the Cleaning/Disinfection Section.
4. Place the endoscope and light cable adapters into the appropriate location within the Mixte Sport Medicine sterilization tray that is compatible with the STERRAD 100S, NX and 100NX sterilization systems. Package trays/instruments with a barrier wrap material in accordance with local procedures, using standardized wrapping techniques such as those described in ANSI/AAMI T9. In the United States, use an FDA-cleared sterilization wrap. Include a STERRAD indicator strip in the wrapped package. Run the STERRAD machine per the manufacturer's instructions.

NOTE: The cleaning solution may contain enzymes. Aluminum-safe alkaline cleaners can be used but can vary in material compatibility overtime based on their formulation. Material compatibility should be confirmed with the detergent manufacturer.

3. Immerse the instruments completely in the detergent solution for a minimum of 15 minutes.
4. Soak for five (5) minutes and then use soft bristle brushes to remove any debris or soil from the instruments while they are submerged. Brush for a minimum of 3 minutes. Ensure that the brushes can access the hard-to-reach areas such as cannulations, cracks, crevices, and threads. Remove all visible debris.
5. Fill a syringe with the detergent and use to flush cannulations, crevices, and threads while submerged.
6. Repeat the brushing step (step 3) and the flushing step (step 4) an additional 2 times.
7. Check for remaining debris and remove as needed.
8. For the sheath only, completely submerge the devices in an ultrasonic bath. Carry out an ultrasonic second soaking bath in a neutral pH detergent solution for 10 minutes using an ultrasonic bath (refer to cleaning agent solution manufacturer's instructions for immersion time and temperature).

NOTE: Ultrasonic cleaning is only effective if the surface to be cleaned is immersed in the cleaning solution. Air pockets will decrease the efficacy of ultrasonic cleaning.

9. Rinse the instruments under cold running tap water for at least three (3) minutes while brushing with a soft bristle brush during the rinse.
10. For the final rinsing processes, soak the devices for a minimum of three (3) minutes in purified (e.g., deionized, reverse osmosis, distilled, or critical water), per AAMI TIR 34. Dry with a lint-free soft cloth.

AUTOMATED CLEANING

Equipment required: Automated washing shall be conducted in a validated washer-disinfecter in compliance to ISO 15883-1 and-2, or to an equivalent standard. Automated washing can be included as part of a validated washing, disinfection, and/or drying cycle in accordance with manufacturer's instructions.

1. Manually clean the instruments by immersion prior to automated cleaning (reference manual cleaning section).
2. Place the endoscope and its light cable adaptors into a separate mesh basket. Ensure that the endoscope and light cable adaptors do not touch each other when placed into the basket. Place a mesh screen over the basket to contain the endoscope and light cable adaptors in the basket during processing.
3. Clean in a validated washer disinfecter using the "INSTRUMENTS" cycle and a pH neutral cleaning agent intended for use in automated cleaning as shown in Table 2. **NOTE:** EnzoI® (1oz/gal) and Prolystica 2X Neutral® (1/8 oz/gal) were used for the validation.

Table 2: Automatic cleaning parameters.

Phase	Recirculation time set points	Water temperature	Detergent type
Pre-wash	2 minutes	Cold tap water	Not applicable
Enzyme wash	1 minute	Hot tap water	Enzymatic detergent
Wash 1	2 minutes	65°C (set point)	Neutral pH instrument cleaner
Rinse 1	15 seconds	Hot tap water	Not applicable
Dry phase	15 minutes	66°C (set point)	Not applicable

Current (A)

Revised (B)

present after cleaning, repeat cleaning procedure.

Thermal Disinfection

Thermal disinfection is recommended to render devices safe for handling prior to steam sterilization. Thermal disinfection should be conducted in a washer-disinfector compliant to ISO 15883-1 and-2, or to an equivalent standard. Thermal disinfection in the washer-disinfector shall be validated to provide an A_0 of at least 600 (e.g., 90°C (194°F) for 1 min). Higher levels of A_0 can be achieved by increasing the exposure time and temperature (e.g., A_0 of 3000 at >90°C (194°F) for 5 min, in accordance with local requirements). Load the device components in the washer-disinfector in accordance with manufacturer's instructions, ensuring that the devices and lumens can drain freely. Lumened devices should be placed in a vertical position. If this is not possible due to space limitations within the washer-disinfector, use an irrigating rack or load carrier with connections designed to ensure an adequate flow of process fluids to the lumen or cannulation of the device if provided.

The following automated cycles are examples of validated cycles (Table 3):

Table 3: Examples of automated validated cycles.

Phase	Recirculation time (mins)	Water temp	Water type
Thermal disinfection	1	> 90°C (194°F)	Critical water (RO, deionized or distilled water)
Thermal disinfection	5	> 90°C (194°F)	Critical water (RO, deionized or distilled water)

STERRAD™ sterilization: endoscope and light adaptors only

CAUTION: The sheaths are not validated for sterilization using STERRAD.

1. Ensure the endoscope and light cable adaptors are disassembled before sterilization.
2. The endoscope and light cable adaptors can be sterilized in the STERRAD™ 100S system, and in the STERRAD™ 100NX, and STERRAD™ NX sterilization systems using the STANDARD cycle.
3. Clean and thoroughly dry the endoscope and light cable adaptors per the section titled "CLEANING, THERMAL DISINFECTION, AND STERILIZATION."
4. Place the endoscope and light cable adaptors into the appropriate location within the Mitek Sport Medicine sterilization tray that is compatible with the STERRAD 100S, NX, and 100NX sterilization systems. Package trays or instruments with a barrier wrap material in accordance with local procedures, using standardized wrapping techniques such as those described in ANSI/AAMI 79. In the United States, use an FDA-cleared sterilization wrap. Include a STERRAD indicator strip in the wrapped package. Run the STERRAD machine per the manufacturer's instructions.

Autoclave (steam) sterilization

CAUTION: Do not use an "immediate use or flash" cycle to sterilize the instruments.

1. Before assembly of the sheath for sterilization, make sure that all O-rings are in place and are not damaged. Remove and replace any O-rings that show any sign of damage or wear. Use a water-soluble lubricant in accordance with the lubricant manufacturer's instructions to lubricate parts as necessary. Remove any excess lubricant. Sterilize the sheath with the stopcocks in the open position.
2. Sterilize the endoscope and light cable adaptors disassembled.
3. If the instruments come with a protective transport sleeve, do not sterilize the endoscopic compatible devices with the protective transport sleeve.